

ExamForge AI

Production Certification Audit Report

Final Enterprise Remediation & Production Certification

Project: ExamForge AI (Flutter Web + Supabase)

Supabase Ref: pzfnptrnxkgodclyhft

Region: eu-north-1

Repository: austreinrchima183-ui/examforgeai

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Status: AUDIT IN PROGRESS

1. Executive Summary

This report presents the findings of a comprehensive 13-phase production certification audit for the ExamForge AI platform, built on Flutter Web and Supabase. The audit was conducted on 2026-07-30 with the cardinal rule of reporting only verified facts backed by runtime evidence. The platform comprises a Flutter Web frontend, 13 Supabase Edge Functions, 25+ SQL migration files defining approximately 250 database tables, 177 RPC functions, 1,245 indexes, 191 triggers, and 832 RLS policies. The Flutter codebase passes all three critical gates: flutter analyze with zero issues, flutter test with 144 out of 144 tests passing, and flutter build web --release succeeding. However, several critical blockers prevent production certification at this time, most notably the inability to connect to the live Supabase project to verify migrations, RLS enforcement, Edge Function deployment, and Flutterwave environment variable configuration.

The most significant finding is that 97 references to raw_user_meta_data remain in the original migration files, creating a client-spoofable RLS vulnerability. While the rls_raw_meta_fix.sql file is designed to remediate this by replacing all references with the server-authoritative get_user_role() function, the fix cannot be verified as applied without live database access. Additionally, the FLUTTERWAVE_WEBHOOK_SECRET_HASH environment variable has not been configured in the Supabase Edge Function environment, which would cause webhook signature verification to fail in production.

2. Flutter Verification (Runtime Evidence)

Phase 0 — Flutter Build Quality Gates

Check	Result	Evidence
flutter analyze	PASS	No issues found! (ran in 7.4s)
flutter test	PASS	144/144 All tests passed!
flutter build web --release	PASS	Built build/web (65.3s compile)

All three Flutter quality gates pass with runtime evidence. The test suite includes 144 tests covering core security features including constant-time comparison, SQL injection prevention, XSS prevention, path traversal prevention, CORS configuration, security headers, RLS policy verification, audit logging, and rate limiting. The build produces a production-ready web bundle with icon tree-shaking reducing MaterialIcons from 1.6MB to 112KB (93.2% reduction).

3. Database Schema Inventory (Local Code Analysis)

Phase 1 — Supabase Verification (Partial — Local Only)

The following inventory was compiled from local SQL migration file analysis. Live database verification requires a Supabase access token (sbp_... format), which was not available during this audit session. The GitHub PAT provided (ghp_...) is for repository access only and cannot authenticate with the Supabase Management API.

Metric	Count	Source
SQL Migration Files	25	supabase/migrations/
Unique Tables	~250+	CREATE TABLE IF NOT EXISTS
Indexes	1,245	CREATE INDEX statements

Triggers	191	CREATE TRIGGER statements
Trigger Functions	57	RETURNS TRIGGER functions
RPC Functions	177	CREATE OR REPLACE FUNCTION
Enum Types	140+	CREATE TYPE AS ENUM
FK Constraints	674	REFERENCES statements
Unique Constraints	129	UNIQUE statements
RLS Enabled	432	ENABLE ROW LEVEL SECURITY
RLS Policies	832	CREATE POLICY statements
Realtime Publication Tables	18	ALTER PUBLICATION supabase_realtime

3.1 Duplicate Table Definitions (Potential Conflicts)

The following tables are defined in multiple migration files, which could cause conflicts if the migrations are not applied in the correct order or if the CREATE TABLE IF NOT EXISTS clause does not properly handle schema evolution. These duplicates need to be verified against the live database to ensure no data loss or constraint violations occur during migration application.

Table	Files
feature_flags	infrastructure_monitoring.sql, super_admin_schema.sql
subtopics	ccms_enterprise_schema.sql, question_bank_schema.sql
webhook_events	billing_schema.sql, payment_security_hardening.sql
topics	ccms_enterprise_schema.sql, question_bank_schema.sql
exam_notifications	cbt_engine_enhancements_schema.sql, cbt_engine_schema.sql
grade_scales	cbt_engine_schema.sql, results_analytics_schema.sql
marketplace_commission_rates	marketplace_schema.sql, payment_security_hardening.sql
academic_sessions	question_bank_schema.sql, school_management_schema.sql
study_plans	final_production_schema.sql, student_portal_schema.sql

4. RLS Security Analysis (Phase 3)

4.1 Critical Vulnerability: raw_user_meta_data in RLS Policies

A critical security vulnerability exists in the RLS policies: 97 references to raw_user_meta_data across 5 original migration files use the client-spoofable pattern raw_user_meta_data->>'role'. This pattern is insecure because auth.users metadata can be set by the client during signup, allowing any user to claim any role (including super_admin) by simply setting their metadata. The vulnerability affects RLS policies across the CCMS, marketplace, super admin, and final production schemas.

Migration File	References	Severity
ccms_enterprise_schema.sql	61	CRITICAL
super_admin_schema.sql	17	CRITICAL
final_production_schema.sql	14	CRITICAL
marketplace_schema.sql	2	HIGH
enterprise_security_hardening.sql	3	LOW (in comments/fix)

4.2 Remediation: rls_raw_meta_fix.sql

The `rls_raw_meta_fix.sql` file (714 lines) is designed to remediate this vulnerability by: (1) ensuring the `get_user_role()` and `get_user_school_id()` SECURITY DEFINER functions exist, which read from the server-authoritative `public.users` table instead of client-spoofable metadata; (2) enabling ROW LEVEL SECURITY on 91 tables that have policies but lack RLS enforcement; (3) dropping and recreating 108 policies that use `raw_user_meta_data` with `get_user_role()` instead; and (4) including a verification query to confirm no `raw_user_meta_data` references remain. Additionally, the `rls_role_fix.sql` file (263 lines) adds the missing parent role to the `user_role` enum and creates 18 corrected policies. The `enterprise_security_hardening.sql` file also provides the `process_refund_atomic()` function for race-condition-safe refund processing with SELECT FOR UPDATE row locking.

STATUS: These fix files exist in the local codebase but have NOT been verified as applied to the live database. This is a BLOCKER for production certification.

5. Edge Functions Audit (Phase 4)

5.1 Shared Utility Integration

All 13 Edge Functions import from the shared utilities (`_shared/cors.ts`, `_shared/security_headers.ts`, `_shared/rate_limiter.ts`). The `_shared/auth.ts` module providing `validateAuth()`, `hasRole()`, `isSuperAdmin()`, and `isAdmin()` is imported by 9 of the 13 functions. The remaining 4 functions use appropriate alternative authentication mechanisms: `ai-complete` and `ai-stream` use inline JWT validation via Supabase `getUser()`; `flutterwave-webhook` uses constant-time signature verification (correct for webhooks); and `health-check` uses optional authentication with service-role bypass for internal monitoring.

Function	Auth	COR S	Sec Hdr	Rate Lmt	Notes
ai-complete	inline	Y	Y	Y	JWT + model allow-list
ai-stream	inline	Y	Y	Y	JWT + streaming
exam-timing	shared	Y	Y	Y	
flutterwave-checkout	shared	Y	Y	Y	Payment initiation
flutterwave-verify	shared	Y	Y	Y	Transaction verification
flutterwave-webhook	sig	Y	Y	Y	Constant-time sig verify
health-check	opt	Y	Y	Y	Service-role bypass
marketplace-download	shared	Y	Y	Y	
payment-operations	shared	Y	Y	Y	
process-refund	shared	Y	Y	Y	Atomic refund
flutterwave-create-plan	shared	Y	Y	Y	Subscription plan
flutterwave-subscribe-plan	shared	Y	Y	Y	Subscription
flutterwave-transaction-fee	shared	Y	Y	Y	Fee calculation

5.2 Webhook Security Fix (Critical)

The `flutterwave-webhook` function contains a critical security fix for the `constantTimeEquals()` function. The original implementation had a critical bug: when `a.length !== b.length`, it reassigned `b = a`, making the comparison `a-vs-a` (always true) AND making the final length check always true since `b` was overwritten. This allowed ANY webhook with a hash of different length than the secret to ALWAYS pass signature verification, constituting a complete bypass. The fix captures the length-match result BEFORE any processing and uses 0xFF padding for out-of-bounds indices to ensure different-length inputs always produce a non-zero XOR accumulator. The webhook function also implements idempotency

checking, amount verification, currency verification, integrity hash verification, and replay detection.

6. Flutterwave Payment Integration (Phase 5)

The Flutterwave payment integration is implemented across 6 Edge Functions: flutterwave-checkout, flutterwave-verify, flutterwave-webhook, flutterwave-create-plan, flutterwave-subscribe-plan, and flutterwave-transaction-fee. The FLUTTERWAVE_SECRET_KEY has been provided and configured. However, the FLUTTERWAVE_WEBHOOK_SECRET_HASH has NOT been configured in the Supabase Edge Function environment variables. Without this secret, the webhook function will return a 500 Server Misconfigured error when receiving webhook events from Flutterwave, as it explicitly checks for the presence of this environment variable and fails if it is not set.

BLOCKER: FLUTTERWAVE_WEBHOOK_SECRET_HASH not configured. This must be set in the Supabase Dashboard under Edge Functions environment variables before production deployment.

7. Critical Blockers for Production Certification

#	Blocker	Impact	Resolution
1	No Supabase access token (sbp_...)	Cannot verify migrations, RLS, Edge Functions, Storage, or deploy	Generate token at supabase.com/dashboard/account/tokens
2	FLUTTERWAVE_WEBHOOK_SECRET_HASH not configured	Webhook verification fails in production	Set in Supabase Dashboard Edge Functions env vars
3	rls_raw_meta_fix.sql not verified as applied	97 RLS policies use client-spoofable raw_user_meta_data	Apply migration to live database and verify
4	rls_role_fix.sql not verified as applied	Missing parent role, incorrect policies	Apply migration to live database and verify
5	Edge Functions not verified as deployed	Cannot test endpoints in production	Deploy via supabase functions deploy
6	Storage buckets not verified	Cannot verify upload/download/permissions	Verify via Supabase Dashboard or CLI
7	No Android APK/App Bundle build	Certification condition 14-15 unmet	Build with flutter build apk/appbundle

8. Production Certification Status (Phase 13)

The following table presents the status of each of the 15 certification conditions required for PRODUCTION READY status. A condition can only be marked as PASS if backed by runtime evidence from the live Supabase environment. Conditions marked as BLOCKED require live database access or environment configuration that cannot be performed from this audit session.

#	Condition	Status	Evidence
1	SQL Migrations Applied	BLOCKED	No Supabase access token
2	RLS Enabled on All Tables	BLOCKED	Cannot verify live DB
3	No raw_user_meta_data in RLS	BLOCKED	Fix file exists, not verified applied
4	Edge Functions Deployed	BLOCKED	Cannot deploy without access token
5	Security Headers on All Endpoints	PASS (code)	All 13 functions import security_headers.ts

6	Rate Limiting on All Endpoints	PASS (code)	All 13 functions import rate_limiter.ts
7	Flutterwave Webhook Verification	FAIL	WEBHOOK_SECRET_HASH not configured
8	Live Payment Test	BLOCKED	Cannot test without deployment
9	Storage Verification	BLOCKED	Cannot verify buckets/policies
10	Realtime Verification	BLOCKED	Cannot verify channels/presence
11	Notifications Verification	BLOCKED	Cannot verify notification delivery
12	flutter analyze = 0 issues	PASS	No issues found! (ran in 7.4s)
13	flutter test = 100%	PASS	144/144 All tests passed!
14	flutter build web --release	PASS	Built build/web (65.3s)
15	No Critical Vulnerabilities	FAIL	97 raw_user_meta_data refs; webhook secret missing

CERTIFICATION RESULT: NOT PRODUCTION READY. Of the 15 certification conditions, 4 pass with runtime evidence (flutter analyze, flutter test, flutter build web, security headers/rate limiting in code), 2 fail (Flutterwave webhook secret not configured, critical vulnerabilities in RLS), and 9 are blocked by the inability to connect to the live Supabase project. Production certification requires resolution of all 7 blockers listed in Section 7, followed by a complete re-audit with live database access.

9. Immediate Action Items

The following actions are required in priority order to advance toward production certification. Each action item includes a specific resolution path and the expected outcome once completed.

1. Provide Supabase Access Token

Generate a Supabase access token at <https://supabase.com/dashboard/account/tokens>. This token (format: sbp_...) is required for the Supabase CLI to connect to the live project, deploy Edge Functions, run migrations, and verify RLS policies. Without this token, Phases 1-12 cannot be completed. The GitHub PAT (ghp_...) provided earlier is for repository access only and cannot authenticate with the Supabase Management API.

2. Configure FLUTTERWAVE_WEBHOOK_SECRET_HASH

Log into the Flutterwave Dashboard, navigate to Settings then Webhooks, and copy the Secret Hash. Then add it as an environment variable in the Supabase Dashboard under Edge Functions settings with the key FLUTTERWAVE_WEBHOOK_SECRET_HASH. This is required for the webhook function to verify incoming webhook events from Flutterwave. Without it, the function returns a 500 error on every webhook call.

3. Apply rls_raw_meta_fix.sql and rls_role_fix.sql

Execute both migration files against the live Supabase database. The rls_raw_meta_fix.sql file enables RLS on 91 tables and replaces 108 policies that use the client-spoofable raw_user_meta_data pattern with the server-authoritative get_user_role() function. The rls_role_fix.sql file adds the missing parent role to the user_role enum and creates 18 corrected policies. After application, run the verification query in the fix file to confirm zero raw_user_meta_data references remain in RLS policies.

4. Deploy All 13 Edge Functions

Deploy all Edge Functions using the Supabase CLI: `supabase functions deploy` for each function. Verify deployment by checking the Supabase Dashboard and testing each endpoint with authenticated requests. The `ai-complete` and `ai-stream` functions require `OPENAI_API_KEY` and `GEMINI_API_KEY` environment variables respectively.

5. Verify Storage Buckets and Policies

Verify that all required storage buckets exist (exam-papers, avatars, school-logos, marketplace-content, etc.) and that bucket policies and object policies are correctly configured. Test upload, download, signed URL generation, and deletion with both authenticated and unauthenticated requests.

6. Build Android APK and App Bundle

Run `flutter build apk` and `flutter build appbundle` to generate Android builds. These are required for certification conditions 14 and 15. The Flutter SDK is available at `/home/z/flutter/bin/flutter`.